BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 6

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Subject Name : Deep Learning

Problem Statement :

Assignment 6. Design and implement a Convolutional Neural Network (CNN) for image classification using the Tomato or Soybean disease dataset.

Code :

```
import kagglehub


# Download latest version

path =
kagglehub.dataset_download("yusufmurtaza01/tomato-leaf-dataset-for-disease-detection")


print("Path to dataset files:", path)

import tensorflow as tf

from tensorflow.keras import layers, models
```

```
import matplotlib.pyplot as plt
import os
import kagglehub

# Re-download/retrieve path to ensure variable is in scope
path = kagglehub.dataset_download("yusufmurtaza01/tomato-leaf-dataset-for-disease-detection")

# Configuration
BATCH_SIZE = 32
IMG_SIZE = (128, 128)

# The previous output shows a 'tomato_yolo_dataset' folder.
# Let's adjust the path to look inside that subfolder if necessary.
adjusted_path = os.path.join(path, 'tomato_yolo_dataset')
if not os.path.exists(adjusted_path):
    adjusted_path = path

# Load datasets
train_ds = tf.keras.utils.image_dataset_from_directory(
    adjusted_path,
```

```
    validation_split=0.2,
    subset="training",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

val_ds = tf.keras.utils.image_dataset_from_directory(
    adjusted_path,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=IMG_SIZE,
    batch_size=BATCH_SIZE
)

class_names = train_ds.class_names
print(f"Detected classes: {class_names}")
# Optimize for performance
AUTOTUNE = tf.data.AUTOTUNE
train_ds = train_ds.cache().prefetch(buffer_size=AUTOTUNE)
```

```
val_ds = val_ds.cache().prefetch(buffer_size=AUTOTUNE)


# Build CNN Model

num_classes = len(class_names)

# If num_classes is 1, we use sigmoid; otherwise softmax.

activation_fn = 'sigmoid' if num_classes == 1 else 'softmax'

loss_fn = 'binary_crossentropy' if num_classes == 1 else
'sparse_categorical_crossentropy'


model = models.Sequential([

    layers.Input(shape=(128, 128, 3)),

    layers.Rescaling(1./255),

    layers.Conv2D(32, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(64, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    layers.Conv2D(128, (3, 3), activation='relu'),

    layers.MaxPooling2D((2, 2)),

    layers.Flatten(),

    layers.Dense(128, activation='relu'),

    layers.Dropout(0.5),

    layers.Dense(num_classes, activation=activation_fn)
```

```
])

model.compile(
    optimizer='adam',
    loss=loss_fn,
    metrics=['accuracy']
)

model.summary()
# Train the model
EPOCHS = 10
history = model.fit(
    train_ds,
    validation_data=val_ds,
    epochs=EPOCHS
)

# Plot results
acc = history.history['accuracy']
val_acc = history.history['val_accuracy']
loss = history.history['loss']
```

```
val_loss = history.history['val_loss']


plt.figure(figsize=(12, 5))
plt.subplot(1, 2, 1)
plt.plot(acc, label='Train Accuracy')
plt.plot(val_acc, label='Val Accuracy')
plt.title('Accuracy')
plt.legend()

plt.subplot(1, 2, 2)
plt.plot(loss, label='Train Loss')
plt.plot(val_loss, label='Val Loss')
plt.title('Loss')
plt.legend()
plt.show()
```

Output :

Model: "sequential"

Layer (type)	Output Shape	Param #
rescaling (Rescaling)	(None, 128, 128, 3)	0
conv2d (Conv2D)	(None, 126, 126, 32)	896
max_pooling2d (MaxPooling2D)	(None, 63, 63, 32)	0
conv2d_1 (Conv2D)	(None, 61, 61, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 30, 30, 64)	0
conv2d_2 (Conv2D)	(None, 28, 28, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 14, 14, 128)	0
flatten (Flatten)	(None, 25088)	0
dense (Dense)	(None, 128)	3,211,392
dropout (Dropout)	(None, 128)	0
dense_1 (Dense)	(None, 2)	258

Total params: 3,304,898 (12.61 MB)
Trainable params: 3,304,898 (12.61 MB)
Non-trainable params: 0 (0.00 B)

```
Found 18158 files belonging to 2 classes.
Using 14527 files for training.
Found 18158 files belonging to 2 classes.
Using 3631 files for validation.
Detected classes: ['images', 'labels']
```

```
Epoch 1/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 24s 40ms/step - accuracy: 0.9987 - loss: 0.0016 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 2/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 26s 17ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 3/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 4/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 5/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 17ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 6/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 7/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 8/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 9/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 8s 18ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
Epoch 10/10
454/454 ━━━━━━━━━━━━━━━━━━━━ 9s 19ms/step - accuracy: 1.0000 - loss: 0.0000e+00 - val_accuracy: 1.0000 - val_loss: 0.0000e+00
```

